Gemma3

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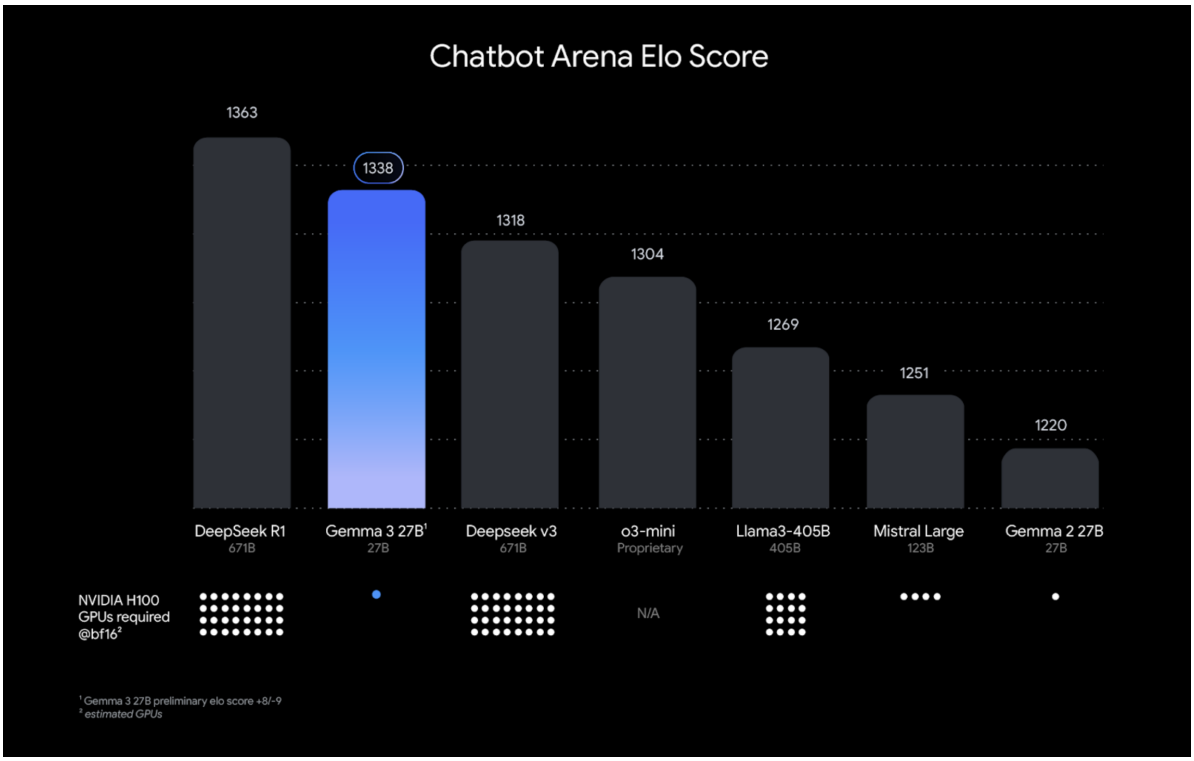
Note: Due to performance limitations, the Raspberry Pi 5 2GB requires the reduced-parameter version.

Gemma is a series of lightweight models built by Google based on Gemini technology. The Gemma 3 model is multimodal (capable of processing text and images), has a 128K context window, and supports over 140 languages.

1. Model Size

Model	Size
gemma3:1b	815MB
gemma3:4b	3.3GB

2. Performance

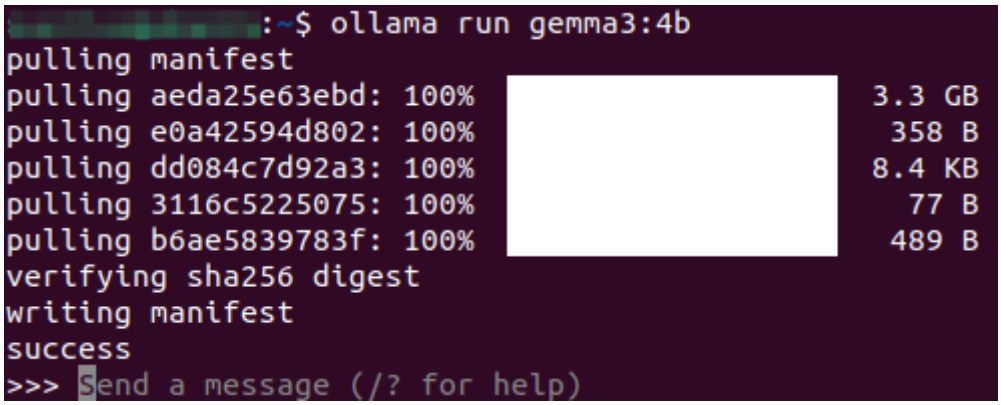


3. Using Gemma3

3.1 Running Gemma3

Use the run command to run the model. If the model is not already downloaded, it will automatically pull the model from the Ollama model library:

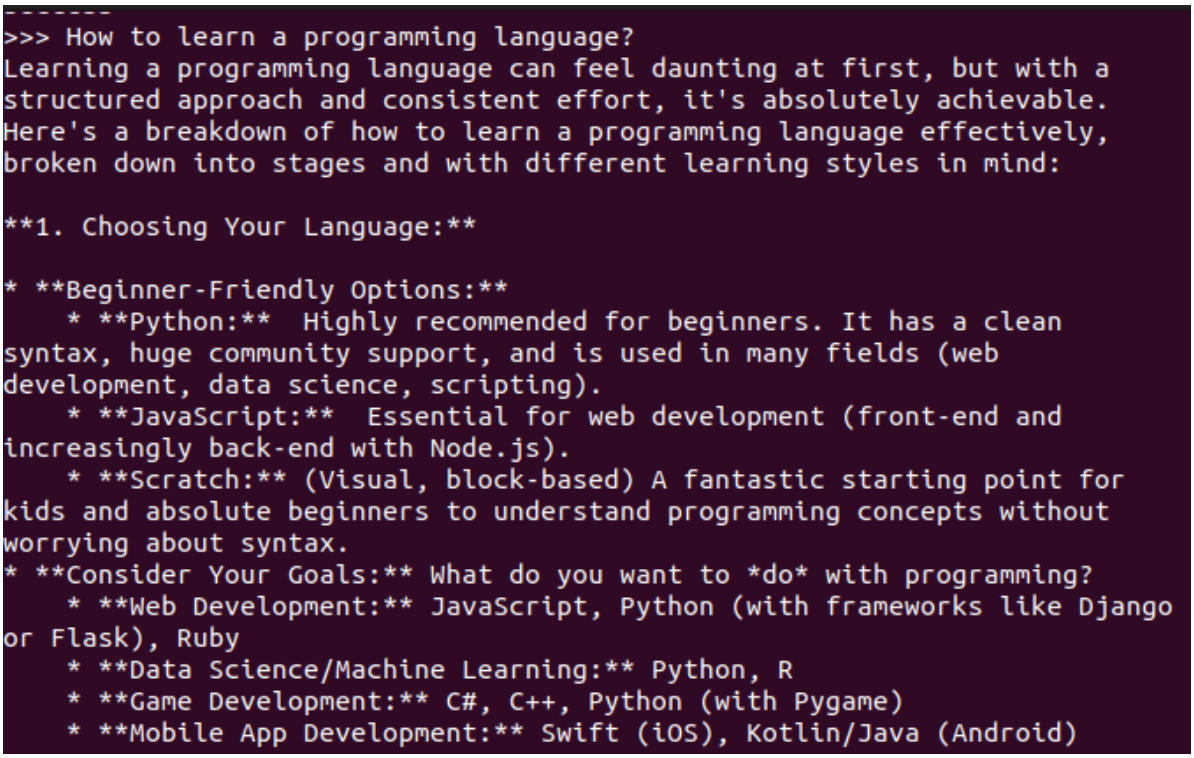
```
ollama run gemma3:4b
```



3.2 Start a conversation

```
How to learn a programming language?
```

Response time depends on hardware configuration, so please be patient!



3.3 Visual Function

```
What do you see in this picture? :./test_pic.png
#Using ": + the image path" in the conversation allows the model to use its
visual function and interpret the information in the image.
```

```
:~$ ollama run gemma3:4b
>>> What do you see in this picture?:./test_pic.png
Added image './test_pic.png'
Here's a breakdown of what I see in the picture:

*   **Main Subject:** A young man is in the foreground, with his eyes
closed and his head tilted back, seemingly taking a deep breath. He's
wearing a white tank top and shorts.
*   **Background:** The background is a cityscape, likely a rooftop with
buildings visible. There is a water bottle next to him.
*   **Other Person:** There’s another man sitting on the right, also
seemingly relaxed.
*   **Sky:** The sky is a vibrant blue with fluffy white clouds.
*   **Color Tone:** The image has a slightly warm, saturated color tone,
giving it a dramatic feel.

This photograph evokes a sense of stillness, perhaps a moment of respite
amidst a busy urban environment. It's a strong, visually compelling image.

>>> Send a message (/? for help)
```

3.4 Ending the Conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

3.5 Chinese Dialogue

If you don't have a Chinese input method, please refer to the Chinese input method switching tutorial.

Chinese Dialogue:

```
:~$ ollama run gemma3:4b
>>> 一个小时是多少分钟？请用中文告诉我
一个小时有 60 分钟。
```

References

> **Ollama**

Official Website: https://ollama.com/

GitHub: https://github.com/ollama/ollama

> Gemma3

Ollama Compatible Model: https://ollama.com/library/gemma3